THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS CHAPTER TEST SET B

Chapter 7: **A Tale of Three Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two angles of a triangle are 35° and 75°. The third angle is: 1

(a) 55° **(b)** 70° **(c)** 145° **(d)** 110°

2. Two angles of a triangle are 90° and 25°. The third angle is: 1

(a) 90° **(b)** 65° **(c)** 0° **(d)** 115°

3. An exterior angle of a triangle is 100°. If one interior opposite angle is 30°, the other is: 1

(a) 80° **(b)** 130° **(c)** 70° **(d)** 110°

4. Sides 2 cm, 3 cm and 6 cm form a triangle? 1

(a) No **(b)** Yes **(c)** Only if equal **(d)** Cannot say

5. Sides 7 cm, 10 cm and 12 cm form a triangle? 1

(a) No **(b)** Yes **(c)** Cannot say **(d)** Only if equal

6. In an isosceles triangle the vertex angle is 80°. Each base angle is: 1

(a) 80° **(b)** 10° **(c)** 100° **(d)** 50°

7. A triangle with one angle greater than 90° is: 1

(a) obtuse-angled **(b)** right-angled **(c)** acute-angled **(d)** equiangular

8. The three medians of a triangle meet at the: 1

(a) circumcentre **(b)** orthocentre **(c)** centroid **(d)** incentre

9. A triangle can have at most ___ obtuse angle(s). 1

(a) one **(b)** three **(c)** two **(d)** none

10. The perimeter of an equilateral triangle of side 7 cm is: 1

(a) 14 cm **(b)** 21 cm **(c)** 49 cm **(d)** 28 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Each angle of an equilateral triangle is how many degrees? Why? 2

12. State the exterior angle property. 2

13. Name the types of triangles based on sides. 2

14. Name the types of triangles based on angles. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How do you construct an altitude of a triangle? 3

16. Find the largest angle of a triangle with angles x, x + 10, x + 20. 3

17. Draw the idea of a median and show that a triangle has three. 3

OR

A right triangle's acute angles differ by 20°. Find them. 3

18. The perimeter of an isosceles triangle is 36 cm and the base 10 cm. Find the equal sides. 3

OR

Is a triangle with sides 3 cm, 4 cm, 5 cm right-angled? Verify with squares. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A triangular plot ABC has ∠A = 50°, ∠B = 60°. 5

(i) Find ∠C.
(ii) Which side is the longest?
(iii) An exterior angle at C.
(iv) The type of triangle.

OR

(i) Find the angles of a triangle if they are in ratio 1 : 2 : 3. 5
(ii) Name the triangle.
(iii) Find the exterior angle at the largest.

20. (i) Show the exterior angle property using a diagram. 5
(ii) If the interior opposite angles are 38° and 52°, find the exterior.
(iii) Find the third angle.

OR

(i) Construct a right triangle with hypotenuse 5 cm and a side 3 cm. 5
(ii) Find the third side.
(iii) Find the area.

END OF QUESTION PAPER — **Total: 40 Marks**

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